Assignment 7

Solutions due by 14 April

Hand calculations

1. (10 points) Consider the numerical solution of $y' = i\omega y$, $y(0) = 1$ using the Explicit Euler method and the second order RK method. Suppose $\omega h = 0.1$ and the differential is integrated for $n = 1000$ time steps (*i.e.* $0 \leq t \leq nh/\omega$). What is the solution after n time steps for each time advancement scheme?

   (a) Comment on the amplitude error between exact and the numerical solution for the two methods. (optional)

2. (15 points) The following scheme has been proposed for solving $y' = f(y)$:

$$y_{i+1} = y_i + \omega_1 k_1 + \omega_2 k_2,$$

   where

$$\begin{aligned} k_1 &= h\ f(y_i) \\ k_0 &= h\ f(y_i + \beta_0 k_1) \\ k_2 &= h\ f(y_i + \beta_1 k_0) \end{aligned}$$

   with h being the step size.
   (a) Determine the coefficients ω_1, ω_2, β_0 and β_1 that would maximize the order of accuracy of the method.

Programming

1. (30 points) Consider the differential equation governing simple harmonic motion: $y(t)$

$$\begin{gathered} y'' + \omega^2 y = 0 \text{ for } t > 0 \\ y(0) = y_0;\ y'(0) = 0. \end{gathered}$$

   Use $y_0 = 1$ and $\omega = 4$.

   (a) If the general form of the exact solution is $y(t) = A\sin(\omega t) + B\cos(\omega t)$, obtain the exact solution to the initial value problem subject to the given conditions
   (b) Re-write the above second order differential equation as a system of first order differential equations.
   (c) Solve this system using *leapfrog* method: $y^{(n+1)} = y^{(n-1)} + 2hf(y^{(n)}, t^{(n)})$ with time step size $h = 0.1$s for $0 \leq t \leq 9$ s. Use Euler explicit method to start the calculation.
   (d) Plot the amplitude errors between the exact and numerical solution at specific time instances where the exact solution attains peak values. (optional)

2. (30 points) Chemical reactions often give rise to stiff systems of coupled rate equations. The time history of a reaction (reaction rate is indicated above the arrow) involving three species A, B and C of the following form:

$$A \xrightarrow{0.04} B \ \text{(slow)} \tag{1}$$

$$B + B \xrightarrow{3\times10^7} C + B \ \text{(veryfast)} \tag{2}$$

$$B + C \xrightarrow{10^4} A + C \ \text{(fast)} \tag{3}$$

is governed by the following rate equations:

$$A: \quad \frac{dy_1}{dt} = -0.04y_1 + 10^4 y_2 y_3 \tag{4}$$

$$B: \quad \frac{dy_2}{dt} = 0.04y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2 \tag{5}$$

$$C: \quad \frac{dy_3}{dt} = 3 \times 10^7 y_2^2 \tag{6}$$

where y_i are the concentration and $\frac{dy_i}{dt}$ are the rate of change of concentration of the respective species. The initial concentrations of the species are $y_1(0) = 0.9, y_2(0) = 0.1$ and $y_3(0) = 0$. Solve this initial value problem using explicit Euler, trapezoidal and fourth order Runge-Kutta methods using a time step sizes of $10^{-3}, 10^{-4}$ and 10^{-5} integrate till $t = 3000$.

(a) Tabulate and compare $y_i(t = 3000), i = 1, 2, 3$ values obtained using different methods with the exact solution in steady state obtained as $y_1 = 0, y_2 = 0$ and $y_3 = 1$.

(b) Plot the variation of species concentration against $log(t)$ for all the methods.

3. (15 points) Solve the following boundary value problem using shooting method

$$\frac{d^2y}{dx^2} = y^2 \tag{7}$$

$$y\Big|_{x=0} = 0; \quad \frac{dy}{dx}\Big|_{x=1} = -0.8;$$

- Take the grid size $\Delta x = 0.01$ and use Euler-Explicit advancement scheme.
- For the first guess, take $\frac{dy}{dx}\Big|_{x=0} = 1$ and for the second, take $\frac{dy}{dx}\Big|_{x=0} = -1$.
- Convergence criterion: **abs** $\left(\frac{dy}{dx}\Big|_{x=1}^{\text{estimate}} + 0.8\right) < 10^{-5}$

(a) Write down: $\frac{dy}{dx}\Big|_{x=0}$ and $\frac{dy}{dx}\Big|_{x=1}$ for first guess, second guess (using above) and two more guesses

(b) For the converged solution, report: $y(x)$ and $\frac{dy}{dx}$ for $x = 0, 0.1, \ldots, 1$.